

CS & IT ENGINEERING

Discrete Mathematics

Set Theory and Algebra

DPP 02 Discussion notes

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#Q. Let A be $\{a, b, c\}$. Let the relation R on A and let $R = \{(b, a), (b, c), (c, a), (c, b), (b, b)\}$.

Which of the following statements about R is/are true?

- ☒ **A** R is neither reflexive nor irreflexive.
- ☒ **B** R is not symmetric.
- ☐ **C** R is transitive.
- ☒ **D** R is not anti-symmetric.

Handwritten notes:

- (a, a) (underlined)
- (a, b) (underlined)
- (c, b) (underlined)
- (b, c) (underlined)
- (c, a) (underlined)
- (b, a) (underlined)
- $a = b$
- $c = b$

#Q. Let A and B be two sets such that $A \times B = \{(1, 1), (2, 2), (3, 1), (3, 2), (1, 2), (1, 4), (2, 1), (2, 4), (3, 4)\}$.

What is the value of $|P(A)| + |P(B)|$? 16

$$A = \{1, 2, 3\} \quad \begin{array}{c} \text{---} \downarrow 2^3 \\ 2^3 \end{array} \quad \begin{array}{c} \downarrow 2^3 \\ 2^3 \end{array} \quad \begin{array}{c} (a, b) \\ \downarrow \quad \downarrow \\ A \quad B \end{array}$$

$$B = \{1, 2, 4\}$$

#Q. The relation R is defined on $Z = \{0, \pm 1, \pm 2, \pm 3 \dots\}$ by xRy if and only if

$(x^2 + y^2) \bmod 4 = 0$. Then R is:

$$(y^2 + x^2) \bmod 4 = 0$$

$$(1^2 + 1^2) \bmod 4 = 0 \quad (1, 1)$$

A Reflexive, Symmetric

B Transitive, Symmetric

C Symmetric

D Reflexive, Symmetric and Transitive

$$(2, 4)$$

$$(4 + 16) \bmod 4 = 20 \bmod 4 = 0$$

$$(4 + 4) \bmod 4$$

$$(4, 2)$$

$$(16 + 4) \bmod 4 = 20 \bmod 4 = 0$$

$$(x, y) \quad (y, z)$$

$$(x, z)$$

#Q. Let R be the relation on the real numbers given by xRy iff $(x - y)^2 < 0$. Then which of the following statement is/are true?

$$(3, 3)$$

$$0^2 < 0 \quad \times$$

$$(x - y)^2 < 0$$

$$\{ 4 \}$$

$\times$

A

R is an equivalence relation.

$\checkmark$ **B**

R is symmetric.

$\checkmark$ **C**

R is asymmetric and anti-symmetric both.

$\checkmark$ **D**

R is transitive.

#Q. Let R be the relation on $\mathbb{R}$ given by xRy if and only if $x < 2y + 1$, then R is

$$\begin{array}{cc} (2, 100) & (100, 2) \\ \checkmark 2 < 201 & 100 < 5 \times \end{array}$$

$$(n, n)$$

$$\Rightarrow n < \underline{2n + 1}$$

A

Reflexive, but not symmetric and not transitive.

B

Reflexive, symmetric, and not transitive.

C

Not Reflexive, not symmetric and transitive.

D

Reflexive, but not symmetric and transitive.

$$\begin{array}{ccc} (5, 3) & (3, 2) & (5, 2) \\ 5 < 7 & 3 < 5 & 5 < 5 \times \end{array}$$

$$\begin{array}{cc} (n, y) & (y, n) \\ n < 2y + 1 & y < 2n + 1 \\ (2, 3) & (3, 2) \\ 2 < 7 & 3 < 5 \end{array}$$

#Q. Let R be the relation on $A = \{1, 2, 3, 4\}$ given by $R = \{(1, 2), (2, 1), (2, 3), (4, 4), (3, 4)\}$. What is the cardinality of the reflexive closure of R ? 8

$(1, 1)$, $(2, 2)$ $(3, 3)$
→

#Q. Let R be the relation on $A = \{1, 2, 3, 4, 5\}$ given by $R = \{(\underline{1, 2}), (\underline{2, 2}), (\underline{2, 4}), (\underline{2, 3}), (\underline{1, 1}), (\underline{3, 3}), (\underline{3, 2})\}$. What is the cardinality of the symmetric closure of R ?

$(2, 1)$ $(4, 2)$

#Q. Let R be the relation on $A = \{1, 2, 3, 4\}$ given by $R = \{(1, 2), (2, 3), (4, 1), (3, 4)\}$. What is the cardinality of the transitive closure of R ? 16

$(1, 3)$

$(4, 2)$

$(2, 4)$

$(4, 4)$

$(1, 4)$

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$(4, 3)$

...

#Q. Suppose that R is the relation on positive real numbers such that xRy if and only if $xy = 1$. Then R satisfies how many of the properties given below?

Reflexive $\times$

Irreflexive $\times$

Symmetric $\checkmark$

Anti-symmetric $\times$

Transitive $\times$

Asymmetric $\times$

$$\begin{array}{l}
 (2, 2) \times \\
 \rightarrow (1, 1) \\
 xy = 1 \\
 yx = 1
 \end{array}
 \quad
 \begin{array}{l}
 (n, n) \checkmark \\
 \rightarrow (3, \frac{1}{3}) \checkmark \quad (\frac{1}{3}, 3) \checkmark \\
 \rightarrow (3, 3) \checkmark
 \end{array}$$

①

#Q. Let R be the relation $\{(a, b) \mid \overbrace{a \neq b}^{\text{check}}\}$ on the set of integers. What is the reflexive closure of R ?

A $\{(a, b) \mid \underline{a = b}\}$

☒ **B** $\{(a, b) \mid a \leq b\}$

☒ **C** $\{(a, b) \mid a \geq b\}$

☒ **D** $\mathbb{Z} \times \mathbb{Z}$



THANK - YOU